

MAJOR CONTRIBUTORS TO THIS REPORT



Anindya Das,
industry manager, Energy
& Power Systems Practice,
Frost & Sullivan



George Mathew,
research analyst,
Netscribes



Manoj Jain,
vice president, Microtek
International Pvt Ltd



Sameer R. Sanghvi,
managing director,
Barr Electronics



T. Anand,
principal consultant,
Knewron



Vikas Chola,
MCU application manager-
North India, Texas
Instruments (India)

per cent efficiency units, is one of the latest trends that has emerged in the market. This helps in conserving energy as well as reducing the carbon footprint."

On a similar note, Das informs, "Today, we see a lot of modular UPS coming in the market. There is also a UPS called Flywheel or Green UPS which uses energy under the flywheel (rotor)." (Flywheel systems store energy kinetically rather than chemically. Instead of dozens of heavy containers of lead plates submerged in sulphuric acid, flywheels use the inertia of a spinning mass to store and regenerate power. Data courtesy: www.power-thru.com)

Introduction of transformerless UPS systems is another emerging area where the efficiency is a little higher. It is used for very specific applications.

Talking about another significant trend, he adds, "Multi-utility products, wherein both lighting and computing power backup requirements are managed, are being manufactured to meet the market needs. Moreover, owing to rising real-estate costs, vendors are also looking to offer products with a higher wattage per square centimetre. Even 5kVA UPS units are being equipped with features that were earlier seen only in high-end systems."

The technology for this market has been developing over the years. Chola shares, "Currently, in the home UPS market, we see the move from traditional square-wave inverters towards the sine-wave inverter technology. Second, the sine-wave inverters are predominantly iron-core transformer types. However, in recent times, development of ferrite-core-

Price trend

There are several players operating in the industry, which has led to a stiff price competition in the market. While unorganised players sell their products at significantly lower prices, products from the organised players are generally sold at premium costs. The difference in prices can be attributed to the quality of the products and the superior after-sales services offered by the bigger brands.

Additionally, high raw material prices and inflationary pressures are making it all the more difficult for players to keep the prices of their products stable.

Courtesy: Netscribes

based high-frequency inverters is taking place. These inverters are smaller in size and more efficient."

The trend with respect to technology is also moving towards solar. Today, high-power tracking systems such as maximum power point tracker (MPPT) based solar inverters are being used. "This is the latest technology shift that is occurring from the typical pulse-width modulation (PWM) inverters. Companies are employing more of this tracking technology to extract the maximum they can from sunlight and convert it into usable power," says Sanghvi.

Another trend is moving towards hybrid inverters. These inverters use solar panels in combination with the main power to charge the battery, thus saving on the electricity cost. These inverters could help in integrating conventional energy sources with regular sources easily and are programmed to use conventional sources in a better way.

Future

While UPS/inverter industry is relatively stable these days, there is need to look out for new trends in advanced electronics such as Internet of Things based systems and smart grid linked sys-

tems. This would help industry to have a different and futuristic perspective and see UPS and inverters beyond just power suppliers, informs Anand.

R&D has become one of the major focus areas of the power backup systems market. For the overall development of the industry, and for players to be able to increase the profitability, it has become imperative for the companies to invest in R&D activities. Making power backup equipment available in smaller sizes would ensure greater revenue, especially from the SME and SOHO segments. Players can also look to foray into e-commerce business models along with doorstep delivery systems.

Another emerging area is that of superior after-sales services. A widespread service centre network will help companies build trust and ensure recommendations from existing customers, thereby adding to their goodwill and brand value. Moreover, cross-selling platforms for low-voltage products via the diffused channels are expected to bring about a greater visibility amongst end users, thus leading to greater revenues. ●

The authors is a senior correspondent at EYF